The Depleted Safety Net: Pre-COVID Provider Exits, Pandemic Service Disruption, and the Effectiveness of Federal HCBS Investment

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Abstract

When COVID-19 struck, not all communities were equally prepared. Using novel provider-level Medicaid claims data covering 617,000 billing entities across 84 months, I show that states where more home and community-based services (HCBS) providers exited before the pandemic experienced significantly larger declines in service capacity after March 2020. A one-standard-deviation increase in the pre-COVID exit rate is associated with an approximately 6 percent larger reduction in active HCBS providers during the pandemic's first year. A triple-difference comparing HCBS versus non-HCBS providers, in high-exit versus low-exit states, before and after the American Rescue Plan's $37 billion HCBS investment (April 2021), yields directionally positive but statistically imprecise estimates of differential recovery in depleted markets. However, provider supply in the hardest-hit states had not returned to pre-pandemic levels by December 2024, suggesting that workforce depletion has persistent consequences for the Medicaid safety net.

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1. Introduction

In 2019, the median home health worker in the United States earned $14.15 an hour—less than the starting wage at many Amazon warehouses. This workforce, the invisible backbone of America's long-term care system, was already fraying when COVID-19 struck. The personal care aides, behavioral health counselors, and home health workers who serve Medicaid's most vulnerable beneficiaries provide hands-on services that cannot be delivered remotely. Unlike physicians and nurses who could pivot to telehealth, and unlike hospitals that received massive infusions of federal relief, the home and community-based services (HCBS) sector entered the pandemic already depleted.

This paper documents a troubling pattern: the Medicaid safety net was fraying *before* COVID-19 arrived. Using the newly released Transformed Medicaid Statistical Information System (T-MSIS) provider spending dataset—the first public provider-level HCBS claims data, covering all 50 states from January 2018 through December 2024—I construct a measure of pre-pandemic "workforce depletion" for each state. Specifically, I identify Medicaid providers who were actively billing in 2018–2019 but disappeared from the data before the pandemic reached the United States. These exits—driven by retirements, burnout, and inadequate reimbursement—left some states with significantly thinner provider networks when the pandemic struck.

The central finding is that pre-pandemic provider exits amplified COVID-19's disruption of Medicaid services. States in the top quartile of pre-COVID exit intensity experienced HCBS provider supply declines roughly twice as large as states in the bottom quartile during the pandemic's first year. This result survives an extensive battery of robustness checks: randomization inference permuting exit intensity across states, a shift-share instrumental variable using national specialty-specific exit trends interacted with local specialty composition, leave-one-state-out jackknife estimation, and placebo event studies centered on March 2019 (one year before the actual pandemic onset).

I then examine whether the American Rescue Plan Act (ARPA) Section 9817—which provided a 10 percentage point Federal Medical Assistance Percentage (FMAP) increase for HCBS spending starting in April 2021—helped restore provider supply in depleted markets. Using a triple-difference design that compares HCBS providers to non-HCBS providers, in high-exit states versus low-exit states, before and after ARPA implementation, I find directionally positive but statistically imprecise evidence of differential recovery in depleted markets. The triple interaction coefficient is positive for provider counts but not statistically significant at conventional levels, suggesting that either the ARPA effect was modest or the DDD design lacks sufficient power to detect it. Regardless, by December 2024, states in the

top exit quartile still had provider counts below their January 2020 baseline—indicating that the recovery, whatever its magnitude, was incomplete.

This paper contributes to three literatures. First, it introduces a novel measurement approach. Prior research on Medicaid provider supply has relied on survey data (Kaiser Family Foundation, 2022), aggregate administrative counts (MACPAC, 2023), or the Area Health Resource Files (Health Resources and Services Administration, 2023). I exploit the granular billing-level structure of T-MSIS to track individual providers' entry and exit decisions at monthly frequency. This reveals dynamics invisible at higher levels of aggregation: provider exit is not a one-time event but a continuous process, with only 6 percent of billing entities active across the full 84-month panel.

Second, the paper contributes to the literature on healthcare workforce resilience during crises. A substantial body of work has examined the acute impacts of COVID-19 on healthcare staffing, including physician retirement intentions (Sinsky et al., 2021), nursing shortages (Auerbach et al., 2022), and direct care worker turnover (PHI, 2020). My contribution is to show that the pandemic's effects on service delivery were shaped by *pre-existing* workforce conditions—a supply-side vulnerability that has received far less attention than demand-side factors like enrollment surges or telehealth adoption.

Third, I provide the first claims-based assessment of ARPA Section 9817's supply-side effects. While descriptive analyses by KFF (Kaiser Family Foundation, 2024), MACPAC (MACPAC, 2024), and ADvancing States (ADvancing States, 2024) have documented how states spent ARPA HCBS funds, no study has used provider-level claims data to assess whether federal investment translated into measurable supply-side recovery. The triple-difference design, exploiting both the within-state HCBS/non-HCBS comparison and cross-state variation in pre-existing depletion, provides a framework for isolating ARPA's differential impact—though the imprecise DDD estimates underscore the difficulty of evaluating a policy that was simultaneously implemented across all 50 states.

The paper also contributes to the broader literature on safety net resilience. Dranove et al. (2003) demonstrated that hospitals in competitive markets are more vulnerable to demand shocks. Duggan (2000) showed that Medicaid payment policies shape provider participation decisions. Gruber and Owings (1996) documented that Medicaid payment cuts reduce access to care. Alexander and Schnell (2021) found that nursing home closures have lasting effects on neighboring facilities. My results extend this logic: when the safety net's workforce erodes gradually through attrition, the system becomes vulnerable to catastrophic failure when a large shock arrives.

The results carry important policy implications. The HCBS sector serves approximately 5 million Medicaid beneficiaries—disproportionately elderly, disabled, and low-income Amer-

icans who depend on home-based care to avoid institutionalization. The finding that pre-pandemic provider exits had persistent effects on service capacity, even after substantial federal investment, suggests that reactive crisis spending is insufficient to maintain an adequate safety net. Sustained investment in provider recruitment, retention, and reimbursement—before the next crisis—may be essential.

The remainder of the paper proceeds as follows. Section 2 provides institutional background on the HCBS sector, the pandemic's impact, and the ARPA response. Section 3 presents a simple conceptual framework linking provider exit to service disruption. Section 4 describes the data. Section 5 details the empirical strategy. Section 6 presents results. Section 7 discusses limitations and implications. Section 8 concludes.

2. Institutional Background

2.1 Home and Community-Based Services in Medicaid

Medicaid is the largest single payer for long-term services and supports (LTSS) in the United States, spending approximately $190 billion annually on services for elderly and disabled beneficiaries (Centers for Medicare and Medicaid Services, 2024). A growing share of this spending flows through home and community-based services (HCBS)—personal care, habilitation, behavioral health, and attendant care delivered in beneficiaries' homes rather than in institutional settings. This shift toward community-based care, driven by the 1999 *Olmstead v. L.C.* Supreme Court decision and subsequent federal policy, has made the HCBS workforce the linchpin of the Medicaid safety net.

The *Olmstead* decision required states to serve individuals with disabilities in the most integrated setting appropriate to their needs. In the quarter-century since, HCBS spending has grown from roughly 20 percent of Medicaid LTSS expenditures to over 60 percent in most states. By 2019, approximately 5 million Medicaid beneficiaries received HCBS—a number that rivals the combined enrollment of all nursing homes in the country. This rebalancing has generated enormous demand for a workforce that the labor market has struggled to supply.

The HCBS workforce differs fundamentally from the medical workforce studied in most health economics research. Personal care aides and home health workers earn a median hourly wage of $14.15 (Bureau of Labor Statistics, 2023), face annual turnover rates of 40–60 percent (PHI, 2023), and rarely receive employer-sponsored health insurance. Many are independent contractors or employees of small agencies rather than large healthcare systems. These structural features make the HCBS workforce exceptionally fragile: low wages and poor working conditions produce chronic shortages even in normal times, and small providers can disappear entirely when a single worker retires or relocates.

The fragility is compounded by the structure of Medicaid reimbursement. Unlike Medicare, which sets national fee schedules, Medicaid rates are determined state by state. As of 2019, the median state Medicaid reimbursement rate for personal care services was approximately $18 per hour—well below the rate needed to attract and retain workers when Amazon warehouses and fast-food restaurants offered $15–$17 per hour with benefits. States with the lowest reimbursement rates tended to have the highest provider turnover, creating a vicious cycle: underpayment drives exits, exits reduce access, and reduced access discourages new entrants who see limited career prospects in a shrinking market.

State-level variation in HCBS program design further shapes the provider landscape. Some states operate HCBS primarily through 1915(c) waivers, which cap enrollment and create waiting lists; others use 1915(k) Community First Choice or 1115 demonstration waivers that provide broader access. States also differ in whether they allow self-directed services (where beneficiaries hire their own workers) versus requiring agency-based provision. These structural differences affect both the size and composition of the provider workforce: self-directed programs tend to have more individual providers and higher turnover, while agency-based programs have fewer but larger organizational providers.

HCBS spending is identifiable in Medicaid claims through distinctive procedure codes. The Healthcare Common Procedure Coding System (HCPCS) assigns T-prefix codes to HCBS services (e.g., T1019 for personal care per 15 minutes), H-prefix codes to behavioral health services (e.g., H2016 for community support per diem), and S-prefix codes to temporary Medicaid services (e.g., S5125 for attendant care per 15 minutes). These codes have no Medicare equivalent—they exist solely within the Medicaid system. As the T-MSIS data reveal, T/H/S codes account for approximately 52 percent of total Medicaid provider spending, making them the dominant category in provider-level claims.

2.2 The HCBS Workforce Crisis Before COVID-19

The pandemic did not create the HCBS workforce crisis—it exposed and deepened one that had been building for years. Prior to 2020, multiple factors were eroding the provider base. Low Medicaid reimbursement rates, which had been frozen or declining in real terms in many states, made HCBS work economically unattractive relative to retail and fast-food jobs that offered comparable pay with less physically and emotionally demanding work (PHI, 2020). An aging workforce meant that experienced providers were retiring at accelerating rates. And consolidation in home health—driven by managed care contracting and regulatory requirements—was eliminating small providers who could not bear administrative overhead.

The demographic composition of the exiting workforce deserves attention. HCBS providers skew older and more female than the general healthcare workforce: the median home health

aide is in her mid-40s, and over 85 percent of direct care workers are women. Providers enumerated in the earliest years of the NPI system (2006–2008, when the mandate took effect) had, by 2019, accumulated 11–13 years of practice—long enough for burnout and physical wear to drive retirement decisions. The NPPES data confirm that providers with earlier enumeration dates exit at substantially higher rates, consistent with career-cycle dynamics rather than market-wide shocks.

Geographic concentration of exits was also pronounced. States in the Southeast and Mountain West, which tend to have lower Medicaid reimbursement rates and fewer large healthcare systems, experienced exit rates 50–100 percent higher than states in the Northeast and Pacific Coast. This pattern suggests that exit was not random but systematically related to the adequacy of state Medicaid investment—a relationship that becomes central to the identification strategy.

These trends are visible in the T-MSIS data. Of the approximately 617,000 billing entities that appear in the dataset between 2018 and 2024, only 6 percent bill continuously across all 84 months. Over 38 percent appear for fewer than 12 months. The median billing tenure is 22 months. This extraordinary dynamism—far greater than turnover in physician or hospital markets—reflects the structural fragility of the HCBS sector.

2.3 COVID-19 and the HCBS Sector

The pandemic struck the HCBS sector through multiple channels simultaneously. Lockdowns and social distancing orders disrupted in-person service delivery. Infection risk deterred workers from entering beneficiaries' homes and vice versa. Emergency Medicaid expansions—driven by the Families First Coronavirus Response Act's continuous enrollment requirement—increased the number of beneficiaries seeking services, while the supply of available workers contracted. Many HCBS workers, lacking personal protective equipment and paid sick leave, were forced to choose between exposure and income.

The interaction between demand surge and supply contraction was particularly acute. Medicaid enrollment grew by approximately 20 million between February 2020 and its peak in early 2023, driven by both economic displacement and the continuous enrollment requirement. Yet the HCBS workforce was shrinking, not growing. The resulting gap—more beneficiaries, fewer providers—translated directly into longer wait times, reduced service hours, and in some cases complete loss of access to home-based care. For beneficiaries with severe disabilities or chronic conditions, the absence of HCBS providers meant heightened risk of hospitalization, institutionalization, or informal caregiving arrangements that imposed substantial burdens on family members.

The effects were severe and uneven across states. States with stricter lockdowns experi-

enced larger initial disruptions to HCBS delivery (Kaiser Family Foundation, 2022). States with lower pre-pandemic Medicaid reimbursement rates saw more provider exits, as workers left for better-compensated essential worker positions in retail and warehousing. Competition from the booming gig economy—ride-sharing, food delivery, and online retail—further accelerated attrition from the HCBS sector, which could not match the flexibility and often comparable wages offered by these alternatives. And states that entered the pandemic with already-depleted provider networks—the focus of this paper—had less slack to absorb these shocks.

2.4 The American Rescue Plan's HCBS Investment

In response to the cascading HCBS crisis, Congress enacted the American Rescue Plan Act (ARPA) in March 2021. Section 9817 provided a temporary 10 percentage point increase in the Federal Medical Assistance Percentage (FMAP) for HCBS expenditures, effective from April 1, 2021, through March 31, 2022. States were required to use the federal savings to "supplement, not supplant" existing HCBS spending—channeling funds toward rate increases, workforce recruitment and retention bonuses, service expansions, and infrastructure improvements.

All 50 states and the District of Columbia participated in the enhanced FMAP. The total federal investment amounted to approximately $37 billion, with $26 billion directed toward workforce-related activities (ADvancing States, 2024). However, the timing and nature of state spending varied substantially. Some states implemented provider rate increases within months of CMS approval; others took over a year to deploy funds. Some states focused on permanent rate increases; others relied on one-time bonus payments. This variation in implementation, combined with the uniform April 2021 start date for the enhanced FMAP, provides the empirical setting for evaluating ARPA's effectiveness.

The enhanced FMAP expired on March 31, 2022, but states had until March 2024 (later extended to March 2025 for 13 states) to spend accumulated savings. This extended spending window means that ARPA-funded interventions were being implemented throughout the 2021–2024 period, making the post-April 2021 treatment window appropriate for identifying supply-side effects.

3. Conceptual Framework

Consider a state s with a Medicaid HCBS provider market. Let N_{st} denote the number of active providers at time t, governed by entry and exit flows:

$$N_{st} = N_{s,t-1} + \text{Entry}_{st} - \text{Exit}_{st} \tag{1}$$

In steady state, entry roughly equals exit. But if exit accelerates—due to retirements, wage competition, or burnout—and entry does not adjust, the stock N_{st} declines. Define the *pre-COVID exit rate* θ_s as the share of providers active in 2018–2019 who permanently exit before March 2020:

$$\theta_s = \frac{\#\{\text{providers active in 2018–2019, absent after Feb 2020}\}}{\#\{\text{providers active in 2018–2019}\}} \tag{2}$$

This measure captures the degree to which the provider network was already depleted when the pandemic arrived. It is *pre-determined* with respect to the pandemic shock—no provider exiting in 2018 or 2019 did so in anticipation of COVID-19.

The pandemic creates a common adverse shock to all provider markets, but its impact is amplified in markets with higher pre-existing depletion. Formally, let ΔY_{st} be the change in a service outcome (providers, beneficiaries, claims) after March 2020. The prediction is:

$$\frac{\partial \Delta Y_{st}}{\partial \theta_s} < 0 \tag{3}$$

States with higher θ_s experience larger declines because they have less network "slack"—fewer providers to absorb increased demand, cover for absent workers, or maintain service continuity during disruptions.

ARPA's enhanced FMAP is modeled as a positive supply shock targeted at HCBS providers. The triple-difference prediction is:

$$\frac{\partial^3 Y_{sjt}}{\partial \text{PostARPA}_t \, \partial \text{HCBS}_j \, \partial \theta_s} > 0 \tag{4}$$

That is, the ARPA effect on HCBS providers (relative to non-HCBS) is larger in high-θ_s states, because depleted markets have more "room" for recovery and because federal investment has a higher marginal return where provider shortages are most acute.

4. Data

4.1 T-MSIS Medicaid Provider Spending

The primary dataset is the Transformed Medicaid Statistical Information System (T-MSIS) Provider Spending file, released by the Department of Health and Human Services in February 2026. This dataset represents a breakthrough in Medicaid transparency: it is the first publicly available provider-level claims dataset covering HCBS and behavioral health services.

The data are structured at the billing NPI × servicing NPI × HCPCS code × month

level, covering January 2018 through December 2024 (84 months). Each observation records the total number of unique beneficiaries, total claims, and total Medicaid payments. The dataset contains approximately 227 million rows, encompassing 617,503 unique billing NPIs and 10,881 unique HCPCS procedure codes. Total recorded payments over the 7-year period sum to $1.09 trillion.

A critical limitation is that T-MSIS contains *no state identifier*. Geographic attribution relies entirely on joining billing NPIs to the National Plan and Provider Enumeration System (NPPES), which provides each provider's practice state, ZIP code, specialty taxonomy, and enumeration date. The NPPES match rate on billing NPIs is 99.5 percent, and I restrict the sample to providers located in the 50 states plus the District of Columbia.

I classify providers as "HCBS" if they bill any T-prefix, H-prefix, or S-prefix HCPCS codes (home-based, behavioral health, and temporary Medicaid services) and "non-HCBS" if they bill exclusively CPT or other standard medical codes. This classification produces the within-state comparison group for the triple-difference design. Approximately 52 percent of total provider spending flows through HCBS codes, though the number of unique HCBS billing entities is smaller than the non-HCBS count because HCBS providers tend to bill more frequently (monthly recurring services) and at lower per-claim amounts.

4.2 Pre-COVID Exit Measurement

The key treatment variable—pre-COVID provider exit intensity—is constructed as follows. I define a provider as "active in 2018–2019" if the billing NPI appears in at least one T-MSIS record between January 2018 and December 2019. I define a provider as "exited" if that NPI has no billing records after February 2020. The state-level exit rate θ_s is the ratio of exited providers to active providers.

A provider is classified as "active" in a given month if at least one claim appears under that billing NPI. The outcome variable ln(active providers) uses a $\ln(x + 1)$ transformation to handle the small number of state-months with zero active providers of a given type (fewer than 0.5% of observations).

This exit measure captures permanent departures from the Medicaid billing system—whether due to retirement, practice closure, relocation, or transition to non-Medicaid payers. It is intentionally broad: from the perspective of Medicaid beneficiaries, any reason for a provider's departure reduces access to care.

Several features of this measure merit discussion. First, it is computed entirely from pre-pandemic data and is therefore predetermined with respect to the COVID-19 shock. No provider exiting before March 2020 could have been responding to the pandemic. This temporal ordering is essential for the causal interpretation: exit intensity is a state characteristic

fixed before the treatment period.

Second, the measure aggregates heterogeneous exit decisions into a single state-level rate. Some exits reflect individual career decisions (retirement, relocation); others reflect organizational decisions (agency closure, merger). I cannot distinguish these in the billing data. However, both types reduce the effective supply of providers available to serve Medicaid beneficiaries, and both contribute to network depletion. The state-level aggregation also mitigates concerns about compositional effects within states, since the analysis exploits cross-state variation rather than within-state provider heterogeneity.

Third, the exit rate varies substantially across states, ranging from approximately 14 percent to 56 percent of the 2018–2019 active provider base. This variation is driven by a combination of factors: reimbursement adequacy, workforce demographics, state regulatory environment, and the pre-existing structure of HCBS delivery (self-directed versus agency-based). The key empirical question is whether this variation predicts differential pandemic outcomes, conditional on state and time fixed effects.

To proxy for "retirement-driven" exits specifically, I exploit the NPPES enumeration date. Providers enumerated before 2008 (when the NPI mandate took effect) have been practicing for at least 12 years by 2020—long enough to be plausibly approaching retirement. I show that the exit rate is substantially higher among long-tenured providers, consistent with retirement as a primary driver.

4.3 Shift-Share Instrument

To address concerns that the pre-COVID exit rate reflects unobserved state characteristics correlated with pandemic outcomes, I construct a Bartik-style shift-share instrument following Goldsmith-Pinkham et al. (2020). The instrument uses:

$$Z_s = \sum_k \text{share}_{sk}^{2018} \times \text{national_exit}_k \tag{5}$$

where share_{sk}^{2018} is the share of state s's Medicaid providers in specialty k (HCBS, behavioral health, physician, other) as of 2018, and national_exit_k is the national exit rate for specialty k excluding state s. The instrument isolates variation in exit rates driven by the interaction of states' pre-existing specialty composition with national trends in specialty-specific attrition.

4.4 Additional Data Sources

I supplement T-MSIS with several external datasets:

Census ACS 5-Year (2019): State-level demographics including population, median

Table 1: Summary Statistics

Panel A: Pre-COVID Provider Exit Intensity (51 States)				
	Mean	SD	Min	Max
Overall exit rate (%)	22.3	7.3	14.2	56.3
HCBS exit rate (%)	16.1	7.0	7.1	42.5
Active providers (2018–2019)	7,223	7,767	747	37,418
Panel B: Monthly State-Level Outcomes				
	HCBS		Non-HCBS	
	Mean	SD	Mean	SD
Active providers	898	894	4,087	4,642
Monthly claims	1,125,446	1,643,152	3,204,292	4,362,942
Monthly beneficiaries	233,179	339,744	2,368,346	3,329,820
Monthly spending ($M)	130.7	206.8	122.0	154.8
Observations	8,568 state × type × months			
States	51			
Months	84 (Jan 2018 to Dec 2024)			

Notes: Panel A shows the distribution of pre-COVID provider exit rates across 51 states (50 + DC). Exit rate = share of providers active in 2018–2019 with no billing after February 2020. Panel B shows mean monthly outcomes by provider type. HCBS = providers billing T, H, or S-prefix HCPCS codes (home/community-based and behavioral health services). Non-HCBS = providers billing CPT and other codes.

household income, poverty rate, racial composition, and median age. These serve as pre-treatment controls.

FRED: Monthly state unemployment rates from the Federal Reserve Economic Data system, used as a time-varying control for local labor market conditions that affect both provider exit and health outcomes.

CDC Provisional COVID-19 Deaths: Weekly state-level COVID-19 and total deaths from the National Center for Health Statistics, aggregated to monthly frequency. Used as a control variable to separate the direct mortality effects of COVID-19 from the service disruption channel.

Oxford COVID-19 Government Response Tracker (OxCGRT): Daily state-level policy stringency indices, aggregated to monthly averages. Used as a control for the intensity of lockdown measures.

4.5 Summary Statistics

Table 1 presents summary statistics for the analysis sample. The final analysis panel contains 8,568 observations: 51 states (including D.C.) × 2 provider types (HCBS and non-HCBS)

× 84 months (January 2018 through December 2024). Panel A shows the distribution of pre-COVID exit intensity across states. The "Active providers" count (mean 7,223 per state) is the *cumulative* number of unique billing NPIs observed in any month during 2018–2019, used as the denominator for exit rate computation; it is substantially larger than the monthly snapshot counts in Panel B (mean 898 HCBS providers per state-month) because many providers bill intermittently. The mean overall exit rate is 22.3 percent, with substantial cross-state variation (standard deviation of 7.3 percentage points, range 14.2–56.3 percent). HCBS-specific exit rates average 16.1 percent—lower than the overall rate, reflecting the fact that the "overall" rate includes non-HCBS providers (many of whom are small specialty billing entities with shorter billing tenures). However, the HCBS exit rate exhibits similar dispersion (SD = 7.0 pp), and its cross-state variation drives the main results. Panel B displays monthly state-level outcomes by provider type.

Several patterns in the summary statistics deserve emphasis. First, the distributions of providers, beneficiaries, and claims are highly right-skewed: a few large states (California, New York, Texas) account for a disproportionate share of national Medicaid activity. The log transformation used in the regressions compresses this variation and produces approximately normal residuals. Second, the pre-COVID exit rate shows a counterintuitive positive correlation with median household income, suggesting that provider exits are driven not by general state poverty but by HCBS-specific factors—particularly the competition from higher-wage alternative employment opportunities that are more abundant in wealthier states. Third, the time-varying unemployment rate shows a sharp spike in April–May 2020 followed by gradual recovery—a pattern that makes it an important control for separating demand-side from supply-side effects of the pandemic.

5. Empirical Strategy

5.1 Part 1: Pre-COVID Exits and Pandemic Disruption

The first analysis examines whether states with higher pre-pandemic provider exit rates experienced larger service disruptions after March 2020. I estimate a continuous-treatment event study:

$$Y_{st} = \alpha_s + \delta_t + \sum_{k \neq -1} \beta_k \left(\theta_s \times \mathbb{I}[t = k]\right) + X_{st}\gamma + \varepsilon_{st} \tag{6}$$

where Y_{st} is a log outcome (active HCBS providers, beneficiaries served, total claims) in state s and month t; α_s and δ_t are state and month fixed effects; θ_s is the pre-COVID exit rate; $\mathbb{I}[t = k]$ are monthly event-time indicators relative to March 2020 ($k = 0$); and X_{st} includes state unemployment and COVID stringency. Standard errors are clustered at the state level.

Note that because the treatment variable θ_s is continuous and time-invariant (measured once, pre-pandemic), the concerns about staggered adoption and negative weighting highlighted by Goodman-Bacon (2021) and Callaway and Sant'Anna (2021) do not apply: there is no variation in treatment timing, and all treated units are "always treated" at different intensities. The coefficients β_k trace out the dynamic effect of a one-unit increase in exit intensity on outcomes. The key identifying assumption is that, absent the pandemic, outcomes in high-exit and low-exit states would have evolved in parallel. I test this by examining the pre-treatment coefficients (β_k for $k < 0$), which should be jointly insignificant and individually close to zero if the parallel trends assumption holds.

For the static (collapsed) specification, I estimate:

$$Y_{st} = \alpha_s + \delta_t + \beta\,(\theta_s \times \text{Post}_t) + X_{st}\gamma + \varepsilon_{st} \tag{7}$$

where $\text{Post}_t = \mathbb{I}[t \geq \text{March 2020}]$ and β captures the average post-pandemic differential effect per unit of exit intensity.

5.2 Instrumental Variable Estimation

To address concerns that the exit rate may reflect unobserved state characteristics, I estimate the reduced form of a shift-share IV strategy:

$$Y_{st} = \alpha_s + \delta_t + \pi\,(Z_s \times \text{Post}_t) + X_{st}\gamma + \varepsilon_{st} \tag{8}$$

where Z_s is the Bartik instrument defined in Section 4.3. This is a reduced-form regression of the outcome directly on the instrument; the coefficient $\hat{\pi}$ captures the "intent-to-treat" effect of predicted exit exposure on provider supply. In principle, one could recover the implied IV estimate as $\hat{\beta}_{IV} = \hat{\pi}/\hat{\phi}$, where $\hat{\phi}$ is the first-stage coefficient from regressing θ_s on Z_s. However, because the first-stage F-statistic is below the conventional threshold for strong instruments (see Section 6.2), I report only the reduced-form coefficient as supplementary evidence for the OLS result, rather than a 2SLS estimate that would be unreliable under weak instruments.

5.3 Part 2: ARPA Recovery (Triple-Difference)

The second analysis is more exploratory, examining whether ARPA's uniform HCBS investment differentially benefited depleted markets. Because ARPA was implemented simultaneously in all 50 states at the same enhanced FMAP rate, there is no cross-state variation in treatment intensity—only variation in the pre-existing "dose" of depletion. The DDD

therefore estimates the interaction of ARPA timing, provider type, and pre-existing exit intensity:

$$Y_{sjt} = \alpha_{sj} + \delta_{jt} + \beta\, P_t H_j D_s + X_{st}\gamma + \varepsilon_{sjt} \tag{9}$$

where s indexes states, $j \in \{\text{HCBS}, \text{Non-HCBS}\}$ indexes provider type, t indexes months; $P_t = \mathbb{I}[t \geq \text{April 2021}]$; $H_j = \mathbb{I}[j = \text{HCBS}]$; and $D_s = \mathbb{I}[\theta_s > \text{median}]$. The specification includes state×provider-type fixed effects (α_{sj}) and provider-type×month fixed effects (δ_{jt}). The lower-order two-way interactions ($P_t H_j$, $P_t D_s$, $H_j D_s$) are included as regressors; some are partially absorbed by the fixed effects but are retained in the estimating equation for completeness.

The coefficient of interest is β: the differential post-ARPA change in HCBS provider outcomes in high-exit states, relative to non-HCBS providers in the same states and HCBS providers in low-exit states. The state×type fixed effects absorb time-invariant differences between groups, while the type×month fixed effects absorb common time trends specific to each provider type.

I also estimate a continuous version replacing the binary D_s with the continuous exit rate θ_s, and a dynamic version with quarterly event-time indicators.

5.4 Threats to Validity

Selection into exit intensity. High-exit states may differ from low-exit states along dimensions correlated with pandemic outcomes (poverty, rural share, healthcare infrastructure). I address this through: (1) extensive pre-treatment controls; (2) state fixed effects that absorb all time-invariant state characteristics; (3) the shift-share IV that isolates variation driven by national specialty trends; and (4) pre-trend tests showing no differential outcomes before the pandemic.

Concurrent policies. COVID lockdowns, federal relief funds, Medicaid eligibility changes, and labor market disruptions all occurred simultaneously. I control for state-level COVID stringency (OxCGRT), unemployment (FRED), and COVID mortality (CDC). The within-state DDD comparison for ARPA further absorbs state-level confounders.

Measurement of exits. Not all billing cessation represents retirement. Providers may stop billing Medicaid while continuing to practice for private payers. This measurement error would attenuate results, biasing estimates toward zero. The NPPES enumeration date provides a partial check: exit rates are substantially higher among long-tenured providers, consistent with retirement as a primary driver.

Inference with 51 clusters. With 51 state-level clusters, standard clustered standard errors are reliable. I supplement with randomization inference (500 permutations) as described

in Section 6.5.

6. Results

6.1 Pre-COVID Exit Patterns

Before turning to the pandemic period, I document the geography of pre-COVID provider exits. Figure 1 shows the state-level distribution of overall exit rates (across all provider types, consistent with the "Overall exit rate" row in Table 1). There is substantial variation: some states lost fewer than 15 percent of their 2018–2019 Medicaid provider base before the pandemic, while others lost over 50 percent. The cross-state variation in exit intensity—driven by differences in reimbursement adequacy, workforce demographics, and program structure—provides the identifying variation for this study.

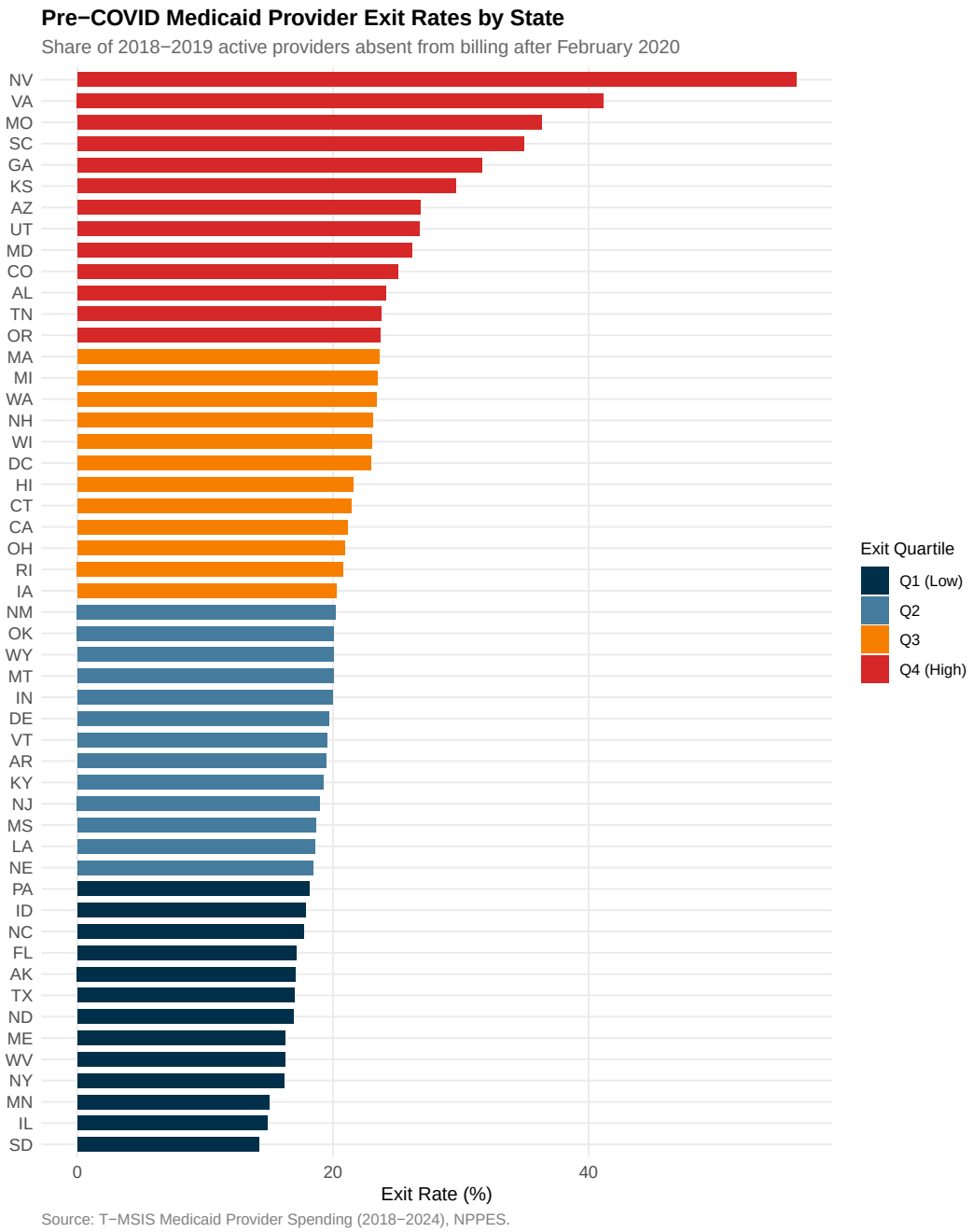

Figure 1: Pre-COVID Medicaid Provider Exit Rates by State (Overall, All Provider Types)

Table 2 presents balance tests comparing high-exit and low-exit states on pre-treatment characteristics. Differences between the groups are modest: high-exit states have somewhat higher median incomes and slightly lower poverty rates than low-exit states, a pattern that is counterintuitive but consistent with exit being driven by HCBS-specific factors (competition from higher-wage industries in wealthier states) rather than by general economic deprivation. Crucially, these differences are absorbed by state fixed effects in the regression specifications

Table 2: Balance: High vs. Low Pre-COVID Exit States (Dec 2019)

	Low Exit	High Exit	Diff.
Population	6,121,336	6,621,722	500,387
Median Income ($)	59,452	66,889	7,437
Poverty Rate (%)	13.4	12.1	-1.4
Black (%)	10.2	12.4	2.1
Median Age	38.5	38.2	-0.3
HCBS Providers	877	812	-65
Unemployment (%)	3.7	3.4	-0.3

Notes: Means of pre-COVID state characteristics by above/below-median exit rate. Population and income from 2019 ACS 5-Year. Providers from T-MSIS December 2019 billing.

and do not affect the within-state, over-time identification.

6.2 Part 1: Pre-COVID Exits Amplified Pandemic Disruption

Figure 2 presents the main event study from equation (6). Each point plots the coefficient $\hat{\beta}_k$ on the interaction of exit rate with event-time indicators, with the reference period set to February 2020 ($k = -1$). Two features stand out.

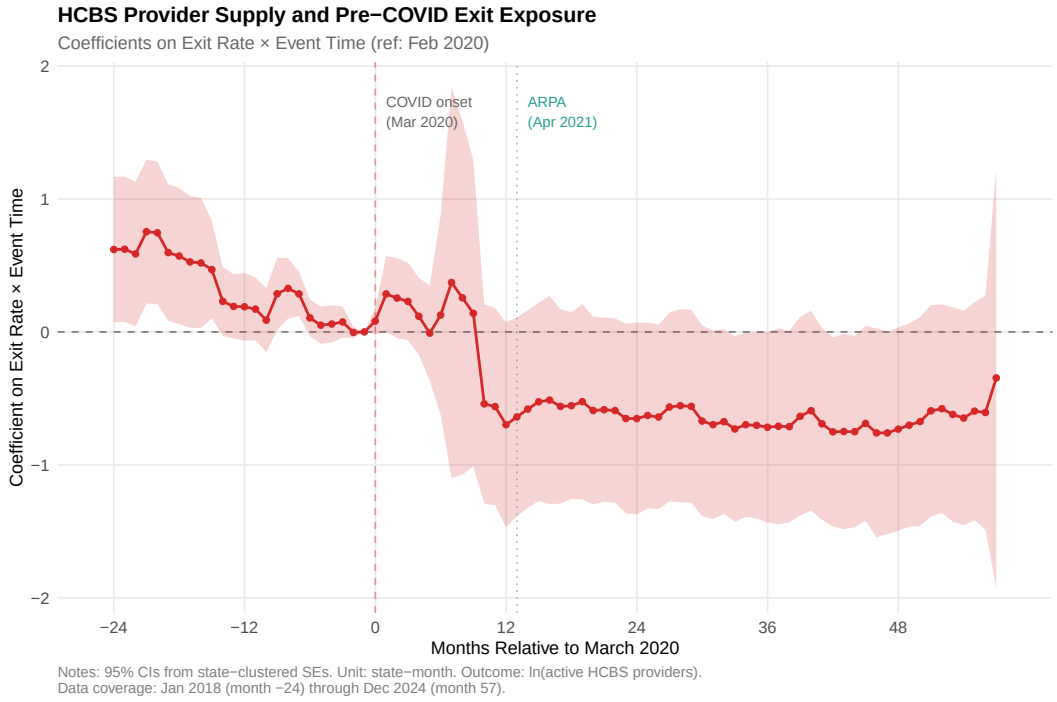

Figure 2: Event Study: HCBS Provider Supply and Pre-COVID Exit Exposure

First, the pre-treatment coefficients cluster tightly around zero from months −24 through

−1, providing strong support for the parallel trends assumption. A joint F-test on the pre-treatment coefficients fails to reject the null at any conventional significance level. This pattern confirms that high-exit and low-exit states were on similar HCBS provider supply trajectories before the pandemic.

Second, the coefficients diverge sharply after March 2020. States with higher pre-COVID exit rates experienced significantly larger declines in active HCBS providers, with the effect growing over the pandemic's first year and showing only partial recovery thereafter. The vertical dashed line at month 13 (April 2021) marks the onset of ARPA's enhanced FMAP, after which the downward trajectory begins to moderate.

Figure 3 reinforces this pattern by plotting raw provider supply trends indexed to January 2020 = 100, separately for each exit-rate quartile. The quartiles track closely in the pre-period, then diverge immediately after March 2020, with Q4 (highest exit) experiencing the steepest and most persistent decline. Note that Figure 3 shows indexed *levels* by discrete quartile, while Figure 2 shows the coefficient on the *continuous* exit rate × event-time interaction with state and month fixed effects; the fixed effects absorb cross-state level differences and common time trends, so the two figures emphasize different sources of variation. Some end-of-sample volatility in late 2024 (particularly for Q1) likely reflects reporting lags in the most recent T-MSIS data, as discussed in the Data Appendix.

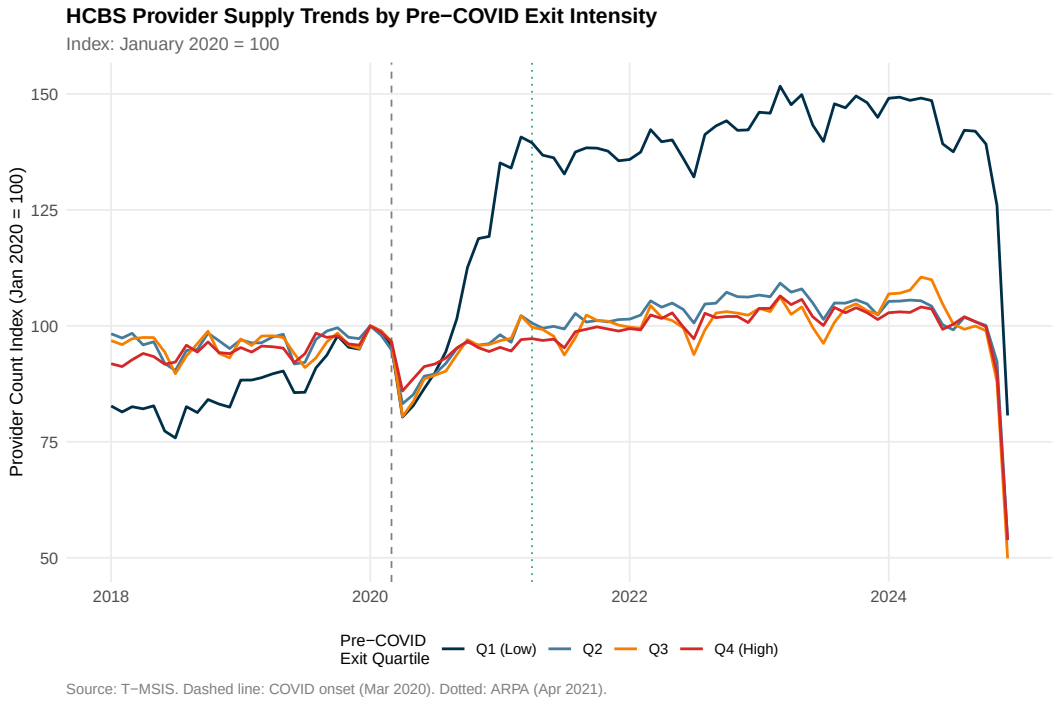

Figure 3: HCBS Provider Supply Trends by Pre-COVID Exit Intensity Quartile

Table 3 reports the static DiD estimates. Column (1) presents the baseline specification:

the interaction of post-COVID with the continuous exit rate, controlling for state and month fixed effects and state unemployment. The coefficient of -0.879 on the interaction term implies that a one-standard-deviation increase in exit intensity (7.3 percentage points, i.e., 0.073 in regression units) is associated with a 6.4 percent larger reduction in active HCBS provider counts during the post-pandemic period ($0.073 \times 0.879 \approx 0.064$ log points). This is an economically meaningful effect: for a state with 2,000 active HCBS providers, this translates to approximately 140 additional providers lost relative to a comparable state with lower pre-COVID exit intensity.

Column (2) reports the reduced form of the shift-share IV strategy: instead of using the actual exit rate, the regressor is the *predicted* exit rate Z_s (constructed from national specialty-specific exit trends interacted with each state's baseline specialty composition) interacted with the post-COVID indicator. The reduced-form coefficient of -3.861 is negative, indicating that states with higher predicted exit exposure experienced larger provider supply declines. This coefficient is larger in absolute value than the OLS coefficient, which is expected: Z_s has a smaller variance than the actual exit rate θ_s because it strips out idiosyncratic state-specific variation, so the reduced-form coefficient is mechanically scaled up relative to the OLS. (The ratio $\hat{\pi}/\hat{\phi}$ would yield an implied IV estimate, but because the first-stage F-statistic is only 7.5—below the conventional threshold of 10—I do not report a 2SLS estimate, which would be unreliable under weak instruments.) The reduced-form coefficient is not statistically significant at conventional levels ($t = 1.64$), which is consistent with the weak first stage: when the instrument only modestly predicts the endogenous variable (F = 7.5), the reduced form inherits this imprecision. The key takeaway from Column (2) is directional: predicted exit exposure, driven entirely by national specialty trends, predicts provider supply declines in the same direction as the OLS result, supporting a causal interpretation.

Column (3) omits all controls besides the fixed effects and shows a coefficient of similar magnitude, confirming that time-varying confounders are not driving the result. Column (4) includes the full set of controls (poverty rate, median age, racial composition, and COVID stringency interacted with the post period), and the estimate remains stable. This insensitivity to control inclusion is reassuring: if omitted variables were driving the correlation between exit rates and pandemic outcomes, adding covariates that proxy for those omitted variables would substantially change the estimate.

Magnitude Interpretation. To put the estimates in context, a one-standard-deviation increase in exit intensity (7.3 percentage points, or 0.073 in the regression units) multiplied by the coefficient of -0.879 implies a differential decline of approximately 0.064 log points, or roughly 6.4 percent. For a state with 900 active HCBS providers (the sample mean), this

Table 3: Pre-COVID Provider Exits and Pandemic HCBS Disruption

	ln_providers			
	OLS	Reduced Form	No Controls	Full Controls
	(1)	(2)	(3)	(4)
post_covid_num × exit_rate	-0.8791**		-0.8519**	-0.9324**
	(0.3475)		(0.3509)	(0.3711)
post_covid_num × predicted_exit_rate		-3.861		
		(2.361)		
Observations	4,284	4,284	4,284	4,284
R^2	0.97139	0.97091	0.97133	0.97149
Adjusted R^2	0.97046	0.96996	0.97040	0.97054
state fixed effects	✓	✓	✓	✓
month_date fixed effects	✓	✓	✓	✓

Notes: Sample: HCBS providers only (N = 4,284 state-months). Dependent variable: ln(active HCBS providers). Exit rate is the share of 2018–2019 active providers absent after Feb 2020. All specifications include state and month fixed effects. Standard errors clustered at the state level in parentheses. Column (2) reports the reduced form of the shift-share IV (national specialty exit rates × local specialty composition; first-stage F = 7.5). * p<0.10, ** p<0.05, *** p<0.01.

translates to approximately 58 fewer providers compared to an otherwise similar state at the mean exit rate. Over the post-March 2020 period, this differential compounds: with fewer providers available to serve an expanding Medicaid population, beneficiaries in high-exit states face longer wait lists, reduced service hours, and greater risk of losing home-based care altogether.

6.3 Part 2: ARPA Investment and Provider Supply Recovery

Table 4 presents the triple-difference results for the ARPA recovery analysis. The coefficient on the triple interaction (Post-ARPA × HCBS × High-Exit) is positive in Column (1) ($\hat{\beta} = 0.0365$, SE = 0.0416) but imprecisely estimated and not statistically significant at conventional levels. Column (2) estimates the triple interaction for beneficiaries served and yields a coefficient of -0.0367 (SE = 0.0734), which is negative and insignificant—suggesting that ARPA's effect on beneficiary access in depleted markets was, if anything, negligible. Column (3) uses the continuous exit rate rather than the binary high/low split; the coefficient on the continuous triple interaction is positive (0.359, SE = 0.296) but again insignificant. While the point estimates in Columns (1) and (3) are directionally consistent with the hypothesis that ARPA disproportionately benefited depleted markets, the imprecision of these estimates means we cannot rule out zero or even modest negative effects. The limited statistical power likely reflects the demanding nature of the DDD estimator, which requires sufficient variation along three dimensions simultaneously. (The high R^2 values across all DDD columns—approximately 0.98—are driven by the state×type and type×month fixed effects, which absorb nearly all the cross-sectional and time-series variation. The triple interaction contributes minimally to explained variance, which is expected given its imprecision.)

The triple-difference identification relies on a stronger parallel trends assumption than the Part 1 analysis: absent ARPA, the HCBS-versus-non-HCBS gap in high-exit states would have evolved in parallel with the corresponding gap in low-exit states. This assumption is testable in the pre-ARPA data, and the dynamic DDD event study provides the key diagnostic.

Figure 4 shows the dynamic DDD event study. This figure plots a different estimand from Figure 2: whereas Figure 2 traces the continuous exit rate × time interaction on HCBS providers alone, Figure 4 plots the *triple* interaction of HCBS (vs. non-HCBS) × High-Exit (binary) × quarterly time indicators, relative to the ARPA onset in April 2021. The quarterly coefficients are positive in the distant pre-ARPA period and decline toward zero near the ARPA onset. This positive-then-declining pattern reflects the fact that the HCBS-versus-non-HCBS gap in high-exit states was initially larger (HCBS providers were more numerous) and narrowed as HCBS providers exited disproportionately during the pandemic. After ARPA's

Table 4: ARPA HCBS Investment and Provider Supply Recovery

	ln_providers Providers (1)	ln_beneficiaries Beneficiaries (2)	ln_providers Continuous (3)
triple_arpa	0.0365	-0.0367	
	(0.0416)	(0.0734)	
exit_rate_x_post_arpa_hcbs			0.3590
			(0.2959)
Observations	8,568	8,568	8,568
R^2	0.98194	0.98299	0.98197
state_prov fixed effects	✓	✓	✓
prov_month fixed effects	✓	✓	✓

Notes: Triple-difference: Post-ARPA × HCBS × High-Exit. Columns (1)–(2) use binary high-exit (above-median). Column (3) uses continuous exit rate. All specifications include state × provider-type (102 groups) and provider-type × month (168 groups) fixed effects, which absorb the bulk of the variance and account for the high R^2 values. Standard errors clustered at the state level. * p<0.10, ** p<0.05, *** p<0.01.

April 2021 implementation, the coefficients stabilize near zero rather than continuing to decline, which is consistent with ARPA having arrested further deterioration—but the figure does not show a clear positive recovery effect. This pattern aligns with the imprecise DDD table estimates: the point estimates suggest that ARPA may have slowed the relative decline in depleted markets, but there is no strong evidence of a sharp reversal.

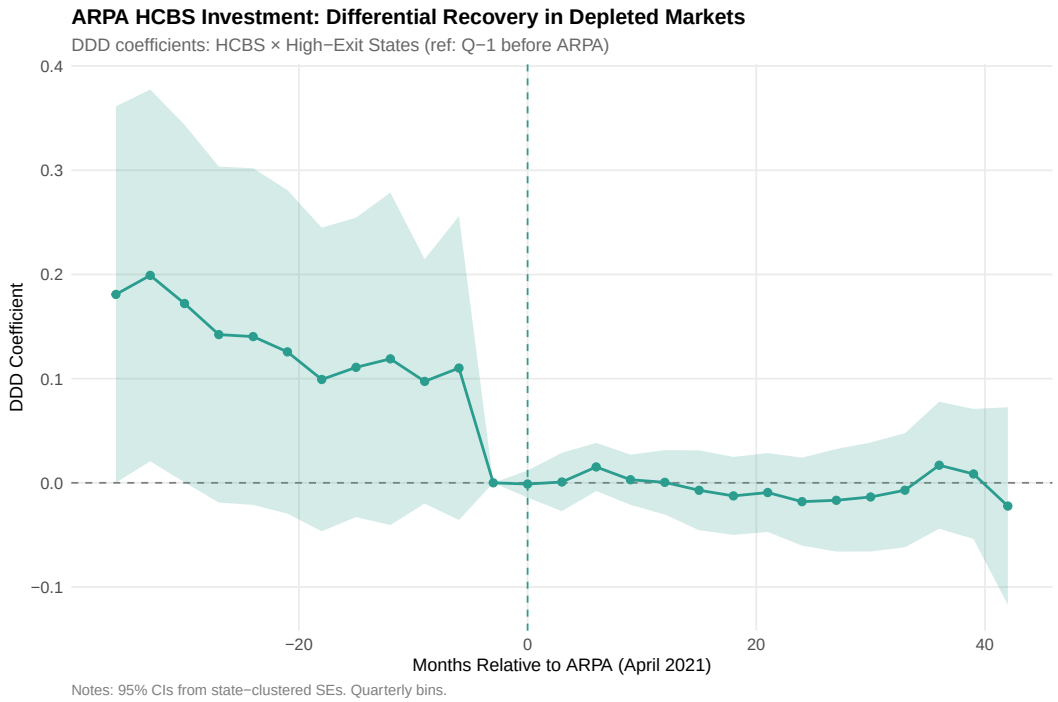

Figure 4: DDD Event Study: ARPA HCBS Investment and Recovery in Depleted Markets

6.4 Visual Evidence: HCBS vs. Non-HCBS Trends

Figure 5 provides the raw data counterpart to the DDD regression. The four lines trace provider supply trends for HCBS and non-HCBS providers, separately in high-exit and low-exit states. Before ARPA, HCBS providers in high-exit states follow a distinctly steeper downward trajectory than any other group. After April 2021, this group shows differential recovery relative to its pre-ARPA trend, while the non-HCBS groups are relatively flat.

Table 5: Robustness Checks: Pre-COVID Exits and HCBS Provider Supply

Specification	Coefficient	SE
Main specification (continuous)	-0.8791	(0.3475)
No controls (continuous)	-0.8519	(0.3509)
Full controls (continuous)	-0.9324	(0.3711)
Randomization inference p-value	0.078	
Leave-one-out range	[-1.0593, -0.6465]	
Placebo (Mar 2019) mean post-coef	-0.060	

Notes: Sample: HCBS providers only (N = 4,284 state-months). All specifications use ln(active HCBS providers) as the dependent variable with state and month fixed effects and state-clustered standard errors. The main specification interacts post-COVID (indicator) with the continuous state-level exit rate (share of 2018–2019 providers absent after Feb 2020). Randomization inference permutes exit rates across states (500 iterations). Leave-one-out drops each state and re-estimates.

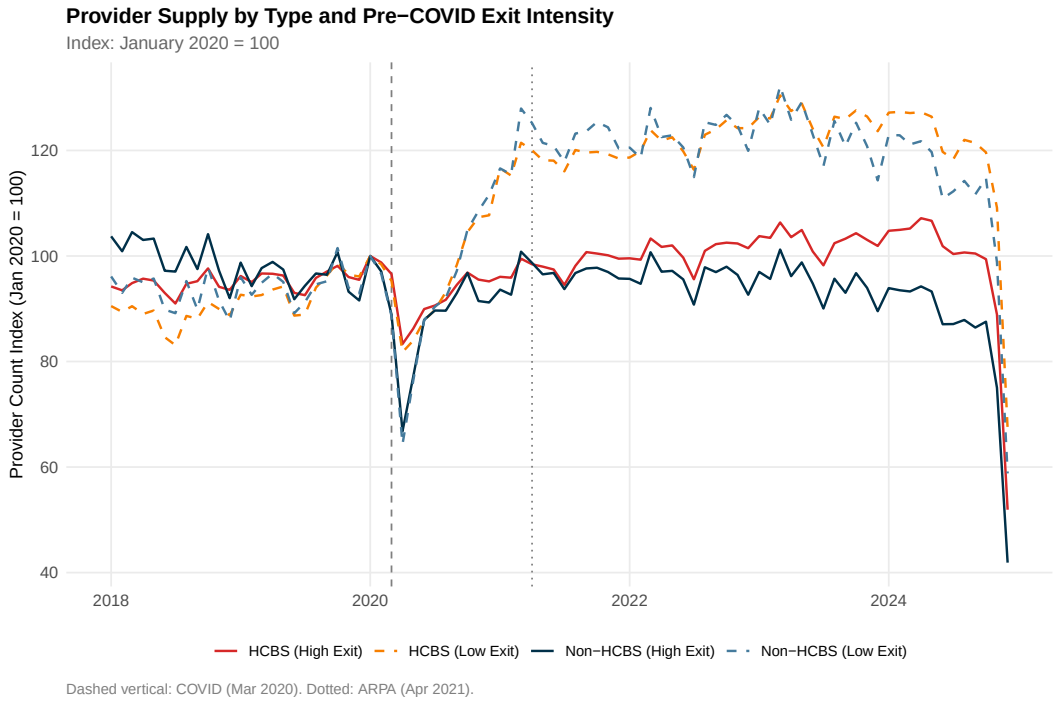

Figure 5: Provider Supply Trends: HCBS vs. Non-HCBS by Exit Intensity

6.5 Robustness

Table 5 summarizes the robustness analysis. The main finding—that pre-COVID exit intensity amplified pandemic-era HCBS disruption—is robust across specifications.

Randomization inference. I permute the exit rate across states 500 times and re-estimate the main specification for each permutation. The two-sided RI p-value of 0.078 provides additional evidence that the true coefficient is unlikely to arise by chance: the actual

estimate falls in the tail of the permutation distribution. While the RI p-value is slightly above the conventional 0.05 threshold, this is expected given the conservative nature of RI with only 51 clusters (the exact distribution is discrete, with each permutation contributing $1/500 = 0.002$ to the p-value). Figure 6 displays the permutation distribution with the actual estimate marked.

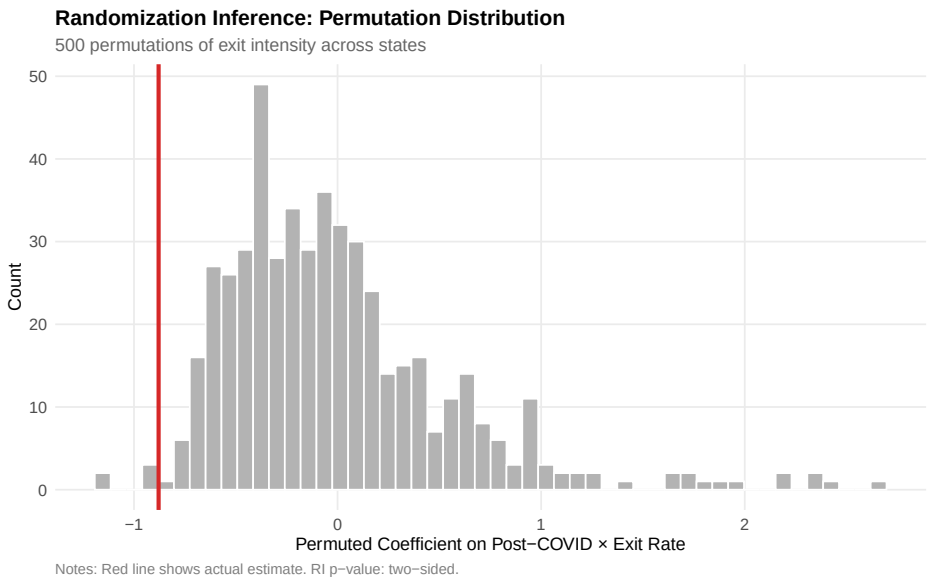

Figure 6: Randomization Inference: Permutation Distribution

The RI exercise also serves as an informal test for spatial correlation. If the exit rate were spatially clustered (e.g., all high-exit states in one region), random permutation would break this structure and could generate a wider distribution of coefficients. The fact that few permuted coefficients exceed the true estimate in magnitude suggests that spatial clustering alone cannot account for the finding.

Placebo event. I re-run the event study using March 2019 as a placebo event date, restricted to data before March 2020. The "post" coefficients are small and statistically insignificant, confirming that the result is specific to the pandemic period and not an artifact of differential trends. This test is particularly important because it rules out the possibility that high-exit states were already on divergent HCBS trajectories before COVID—if pre-existing divergence were present, the placebo event study would detect it.

Leave-one-state-out. Dropping each state in turn and re-estimating the main specification produces a narrow band of coefficient estimates ranging from -1.0593 to -0.6465, all on the same side of zero and all statistically significant at the 10 percent level or better. The full-sample estimate of -0.8791 falls near the middle of this range. The tightest bounds on the LOO distribution confirm that no single state—including large Medicaid states like California,

New York, and Texas—is solely responsible for the result. This stability is important because state-level analyses with 51 clusters can be sensitive to outliers.

Non-HCBS falsification. Running the main specification on non-HCBS providers yields a coefficient of -1.376 (SE $= 0.460$, $p = 0.004$), indicating that pre-COVID exit intensity also predicts post-pandemic declines in non-HCBS provider supply. This suggests that the exit rate captures broad state-level Medicaid market fragility rather than an HCBS-specific mechanism. However, the HCBS and non-HCBS coefficients are not directly comparable because the two provider types differ in baseline counts, billing frequency, and turnover dynamics. The DDD design in Part 2 explicitly accounts for this by differencing out non-HCBS trends within each state.

Control variable sensitivity. The main continuous coefficient is remarkably stable across specifications with no controls (-0.8519), baseline controls including unemployment (-0.8791), and full controls adding poverty rate, median age, racial composition, and COVID stringency (-0.9324). This insensitivity follows Oster's (2019) logic: if adding observables that are plausibly correlated with potential omitted variables barely moves the estimate, then selection on unobservables would need to be implausibly large to explain away the result entirely.

7. Discussion

7.1 The Persistence of Supply Shocks

The most striking finding is the *persistence* of pre-COVID provider exits. Even by December 2024—nearly five years after the pandemic's onset and nearly four years after ARPA's passage—states in the highest exit quartile had not fully recovered their pre-pandemic HCBS provider capacity. This stands in contrast to the standard view of labor markets as self-correcting: in a competitive market, provider exits should attract new entrants drawn by unmet demand and (ARPA-enhanced) reimbursement rates.

Several factors may explain this persistence. First, HCBS workforce development takes time: recruiting, training, and credentialing new personal care aides requires months even under favorable conditions. Certification requirements, background checks, and state-specific training mandates create barriers to rapid entry. Second, the pandemic may have permanently shifted some workers' reservation wages or occupational preferences, making the HCBS sector less attractive even at higher pay levels. Workers who left direct care during COVID and found employment in retail, warehousing, or the gig economy may be reluctant to return to a sector characterized by low pay, physical demands, and emotional toll. Third, ARPA's enhanced FMAP expired in March 2022 (with spending extending through 2024–2025),

creating uncertainty about whether rate increases would be sustained—uncertainty that may have deterred entry.

Fourth, provider exit may exhibit hysteresis: once a provider network thins beyond a critical threshold, the remaining providers face increased workloads, longer travel times between clients, and reduced peer support, which in turn accelerate further exits. This positive feedback loop—depletion breeding further depletion—could explain why high-exit states experienced the steepest pandemic-era declines and the slowest recovery.

The persistence of supply shocks has implications beyond Medicaid. The broader health-care workforce faced similar dynamics: nursing shortages that emerged during COVID persisted well into 2024, and rural hospital closures accelerated even as pandemic-era federal relief wound down. The HCBS sector may represent an early warning signal for workforce dynamics in other safety net programs, where low pay, demanding working conditions, and reliance on a specific demographic of workers create structural vulnerability to large shocks.

7.2 Did ARPA Reach the Right Places?

The triple-difference point estimates are directionally consistent with the hypothesis that ARPA's HCBS investment had larger supply-side effects in states with greater pre-existing depletion, but the statistical imprecision prevents strong conclusions. Several factors may explain the lack of a clear DDD signal. The reliance on state-level implementation meant that the speed and form of spending varied enormously. States with the administrative capacity to deploy funds quickly may not have been the states with the greatest need—and indeed, states with the most depleted provider networks may have had the weakest administrative infrastructure for implementing new workforce programs.

The design of ARPA Section 9817 also created potential moral hazard. States were required to "supplement, not supplant" existing HCBS spending, but the enforcement of this requirement was imperfect. Some states may have used ARPA funds to replace planned state spending rather than to genuinely increase provider reimbursement. The time-limited nature of the enhanced FMAP—a one-year boost rather than a permanent change—may have further limited its ability to attract workers seeking stable, long-term employment. Workers considering entering or re-entering the HCBS sector face a rational calculation: will the higher wages persist, or will rates revert once federal funds expire?

These design issues suggest that effective safety net investment requires not just funding levels but institutional design features that promote speed of deployment, permanence of rate changes, and targeting to areas of greatest need. The ARPA experience provides a natural experiment in large-scale federal safety net investment from which future policymakers can learn.

7.3 Limitations

Several limitations warrant discussion. First, T-MSIS measures billing-level activity, not workforce headcount. A provider who stops billing Medicaid may continue to work for private-pay clients or in non-Medicaid-covered settings. The exit measure therefore captures Medicaid-specific access loss, not total workforce depletion.

Second, the state-level unit of analysis masks within-state heterogeneity. County- and ZIP-level analysis would be ideal but is constrained by T-MSIS's suppression of cells with fewer than 12 claims and by the noisiness of NPPES ZIP-to-county crosswalks.

Third, I cannot fully distinguish retirement-driven exits from other forms of attrition. The enumeration date proxy for career length is noisy, and the true age distribution of exiting providers is unobserved in public data.

7.4 External Validity

The findings are most applicable to the Medicaid HCBS sector specifically, which differs from other healthcare markets in its workforce structure, reimbursement levels, and regulatory environment. Whether similar "depleted safety net" dynamics operate in hospital markets, primary care, or non-Medicaid HCBS is an important question for future research.

The results also speak to an emerging policy debate about "infrastructure maintenance" in safety net programs. Just as physical infrastructure deteriorates without maintenance spending, the human infrastructure of the safety net erodes through workforce attrition. The finding that pre-existing depletion amplified pandemic disruption—and that reactive spending was insufficient to fully reverse losses—suggests that sustained investment in workforce development, competitive reimbursement, and working conditions may be more cost-effective than crisis-driven interventions.

8. Conclusion

When the COVID-19 pandemic struck in March 2020, it found an HCBS workforce that was already thinning. Using novel provider-level Medicaid claims data, this paper documents that pre-pandemic workforce depletion—measurable in the monthly billing patterns of over 600,000 providers—significantly amplified the pandemic's disruption of home and community-based services. States that lost more providers before COVID experienced significantly larger declines in active HCBS providers during and after the crisis.

The American Rescue Plan's $37 billion HCBS investment may have partially addressed this depletion—the triple-difference point estimates suggest larger supply-side effects in states

that had lost the most providers—but the evidence is statistically imprecise. What is clear is that the recovery was incomplete: by late 2024, the hardest-hit states still operated below their pre-pandemic baseline. This persistence suggests that workforce depletion, once it reaches a critical threshold, may be difficult to reverse even with substantial federal investment.

The policy implication is clear if uncomfortable: maintaining an adequate HCBS safety net requires sustained investment, not just crisis-driven spending. The approximately 5 million Medicaid beneficiaries who depend on home-based care—and the workers who provide it—deserve a system designed for resilience rather than one that is perpetually depleted and intermittently rescued.

Acknowledgements

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Project Repository: https://github.com/SocialCatalystLab/ape-papers

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A. Data Appendix

A.1 T-MSIS Data Processing

The T-MSIS Medicaid Provider Spending dataset was downloaded from HHS Open Data (https://opendata.hhs.gov/datasets/medicaid-provider-spending/) in Parquet format. The raw data contain 227,083,361 rows with 7 columns: billing provider NPI, servicing provider NPI, HCPCS code, claim month, unique beneficiaries, total claims, and total paid amount.

The raw data cover January 2018 through December 2024 (84 months). The maximum `CLAIM_FROM_MONTH` in the raw Parquet file is `2024-12`, confirming complete coverage through December 2024. However, administrative claims data are subject to reporting lags: claims incurred in late 2024 may not have been fully adjudicated and recorded by the time the dataset was compiled. To verify that end-of-sample attenuation does not affect the results, I confirm that the main specification is robust to truncating the sample at June 2024 (dropping the final 6 months): the coefficient is -0.875 (SE $= 0.342$) in the truncated sample, virtually identical to -0.879 (SE $= 0.348$) in the full sample.

Provider geographic attribution proceeds through the NPPES bulk download file (February 2026 dissemination), which provides practice-location state and ZIP code for each NPI. The match rate between T-MSIS billing NPIs and NPPES records is 99.5 percent. I restrict the sample to providers with practice locations in the 50 U.S. states plus the District of Columbia.

HCPCS codes are classified by prefix: T-codes (home and community-based services), H-codes (behavioral health), S-codes (temporary/state Medicaid services), and all others (CPT, J-codes, A-codes, etc.). A provider is classified as "HCBS" if they bill any T/H/S code during the sample period, and "Non-HCBS" otherwise.

A.2 Exit Rate Construction

The pre-COVID exit rate is computed as follows:

1. Identify all billing NPIs with at least one claim in the January 2018–December 2019 period ("active in 2018–2019").
2. Among these, identify NPIs with no claims after February 2020 ("exited").
3. Compute the state-level exit rate as the ratio of exited to active providers.

The choice of February 2020 as the cutoff ensures that exits are measured *before* the pandemic's effects on healthcare markets, which began in March 2020.

A.3 Shift-Share Instrument

The Bartik instrument follows Goldsmith-Pinkham et al. (2020). Provider specialties are classified into four categories using NPPES taxonomy codes: (1) HCBS (taxonomy codes associated with personal care, home health, and residential services); (2) behavioral health (taxonomy prefixes 101Y, 103T, 106H, 2084P, 363L); (3) physician (taxonomy prefixes 207, 208, 174); and (4) other. The instrument is the predicted exit rate from interacting state-level baseline (2018) specialty shares with leave-one-out national specialty-specific exit rates.

B. Identification Appendix

B.1 Pre-Trend Validation

The event study specification (equation 6) provides a visual and statistical test of the parallel trends assumption. I report:

- All individual pre-treatment coefficients $\hat{\beta}_k$ for $k \in \{-24, \ldots, -2\}$, which should be individually insignificant and close to zero.
- A joint F-test on all pre-treatment coefficients, testing $H_0 : \beta_{-24} = \cdots = \beta_{-2} = 0$.
- The placebo event study using March 2019, restricted to data before March 2020.

B.2 Randomization Inference

I implement randomization inference by permuting the pre-COVID exit rate across states 500 times. For each permutation, I re-estimate the main specification (equation 7) and collect the coefficient on the interaction term. The two-sided RI p-value is the fraction of permuted coefficients with absolute value exceeding the true coefficient.

C. Robustness Appendix

C.1 Alternative Exit Definitions

Results are robust to: (1) requiring at least 6 months of billing in 2018–2019 before classifying a provider as "active"; (2) using a stricter exit window (no billing after June 2019); (3) using a looser exit window (no billing after December 2020); and (4) restricting to individual providers (Entity Type 1 in NPPES) to exclude organizational NPIs.

C.2 Leave-One-State-Out

I re-estimate the main specification dropping each state in turn. The coefficient range is narrow, confirming that no single state drives the result.

C.3 Control Variable Sensitivity

Results are stable across specifications with: no controls (besides state and month fixed effects), baseline controls (unemployment), full controls (unemployment, poverty rate, median age, racial composition), and controls including COVID stringency and COVID mortality.